MecHand: Freehand Chemical Mechanism Drawing with Sketch Recognition and Direct Export to LaTeX

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Abstract

Background: Reaction mechanisms are central to chemical education and research, yet existing tools for drawing them impose a steep learning curve or require expensive licences. Freehand sketching on paper remains the preferred medium for most chemists, but paper sketches cannot be directly integrated into digital manuscripts.

Results: We present **MecHand**, an open-source web application built on the Ketcher chemical drawing platform that accepts freehand input from stylus, trackpad or mouse and converts sketched mechanisms into clean, editable chemical diagrams. An online sketch recogniser running in the browser reads the pen strokes directly: geometric rules separate bonds, atom labels, charges and curved electron-flow arrows, atom labels are classified with the $P point-cloud recogniser, and every curved arrow is anchored to an atom or bond and checked for consistent electron flow. A "consolidating" interface presents the recognised structure to the user for confirmation or correction before it is committed to the canvas. MecHand exports diagrams, including curved arrows, directly to TikZ and `chemfig` for LaTeX, and to ChemDraw (CDXML) and MDL formats. A context-sensitive help system provides tool descriptions calibrated for both expert chemists and undergraduate students.

Conclusions: MecHand fills a gap between the informality of hand sketching and the rigidity of graphical editors, lowering the barrier to producing publication-quality mechanism figures. The tool is freely available at https://github.com/JordiVillaFreixa/MecHand and runs entirely in the browser without installation, with an optional self-hosted DECIMER server for image-based structure recognition.

Keywords: reaction mechanism, chemical drawing, sketch recognition, electron-flow arrows, chemfig, TikZ, open source

1 Background

Reaction mechanisms are routinely drawn by hand on paper or whiteboards during teaching and research discussions. Translating these sketches into digital manuscript-quality figures currently requires either expensive commercial software (e.g. ChemDraw, Marvin) or manual reconstruction in programmatic tools such as `chemfig` [1] or TikZ [2].

2 Implementation

2.1 Architecture overview

MecHand is structured as a three-layer system (Figure **??**):

1. **Frontend** — the Ketcher editor [3], used as a pinned npm dependency rather than a fork, augmented with a freehand ink layer, the consolidating interface, a context-sensitive help system, mechanism templates and chemfig/TikZ export modules.
2. **Online sketch recogniser** — a TypeScript module running in the browser that turns pointer-event strokes into a molecular graph with curved electron-flow arrows, using ShortStraw corner finding [4] and the $P point-cloud recogniser [5] for atom labels and charges (Sections 2.2 and 2.4).
3. **Optional recognition server** — a FastAPI service wrapping DECIMER [6, 7] as an alternative, image-based engine for the structures.

2.2 Freehand input and stroke classification

2.3 The consolidating paradigm

2.4 Mechanism arrow detection

2.5 Export to chemfig and TikZ

2.6 Context-sensitive help system

3 Results and discussion

3.1 Benchmark on hand-drawn structures

We evaluated the recognition accuracy of MecHand on the DECIMER hand-drawn molecule dataset [8], which contains X structures scanned from student notebooks and published manuscripts.

3.2 User evaluation

4 Conclusions

5 Availability and requirements

- **Project name:** MecHand
- **Project home page:** https://jordivillafreixa.github.io/MecHand/ (source: https://github.com/JordiVillaFreixa/MecHand)
- **Operating system(s):** Platform independent (web application)
- **Programming language:** TypeScript (web application); Python (optional recognition server)
- **Other requirements:** A current web browser (Chrome, Edge, Firefox or Safari; tested automatically with Chromium, Firefox and WebKit); optional Python 3.10–3.13 for the DECIMER recognition server.
- **License:** MIT (MecHand additions); Apache 2.0 (Ketcher upstream)
- **Any restrictions to use by non-academics:** None

Abbreviations

SVG: Scalable Vector Graphics; SMILES: Simplified Molecular-Input Line-Entry System; UX: user experience; API: application programming interface.

Declarations

Competing interests

The authors declare that they have no competing interests.

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Authors' contributions

References

[1] Tellechea C.: The `chemfig` package. Available from: https://ctan.org/pkg/chemfig.

[2] Tantau T.: The TikZ and PGF packages. Available from: https://ctan.org/pkg/pgf.

[3] EPAM Systems.: Ketcher: A Web-Based Chemical Structure Editor. Available from: https://github.com/epam/ketcher.

[4] Wolin A, Eoff B, Hammond T. ShortStraw: A Simple and Effective Corner Finder for Polylines. In: Eurographics Workshop on Sketch-Based Interfaces and Modeling (SBIM); 2008. p. 33–40.

[5] Vatavu RD, Anthony L, Wobbrock JO. Gestures as point clouds: a $P recognizer for user interface prototypes. In: Proceedings of the 14th ACM International Conference on Multimodal Interaction (ICMI '12); 2012. p. 273–280.

[6] Rajan K, Brinkhaus HO, Agea MI, Zielesny A, Steinbeck C. DECIMER.ai: an open platform for automated optical chemical structure identification, segmentation and recognition in scientific publications. Nature Communications. 2023;14:5045. https://doi.org/10.1038/s41467-023-40782-0.

[7] Brinkhaus HO, Rajan K, Zielesny A, Steinbeck C. Advancements in hand-drawn chemical structure recognition through an enhanced DECIMER architecture. Journal of Cheminformatics. 2024;16:78. https://doi.org/10.1186/s13321-024-00872-7.

[8] Rajan K, Zielesny A, Steinbeck C. DECIMER 2.0: Deep Learning for Chemical Image Recognition using Transformers. Journal of Cheminformatics. 2023;15:17. https://doi.org/10.1186/s13321-023-00703-1.